5 Probability 2

Two events are **independent** if the occurence of one does not in any way inform us about the probability that the other will also occur.

Example: E_1=The first die rall is 3, and E_2=The second die roll is 3.

The **multiplication rule** states that when two events are independent, then the probability that they both occur is the product of their probabilities. That is, if E_1 and E_2 are independent, then

$$P(E_1 \text{ and } E_2) = P(E_1)P(E_2)$$

5.1 Conditional probability and Bayes' theorem

Conditional probability is the probability of an event given that another event occurs.

Bayes' theorem

P(A and B) = P(A|B) P(B)

$$\boxed{P(A \mid B) = \frac{P(A \text{ and } B)}{P(B)}}$$

The *general multiplication rule*

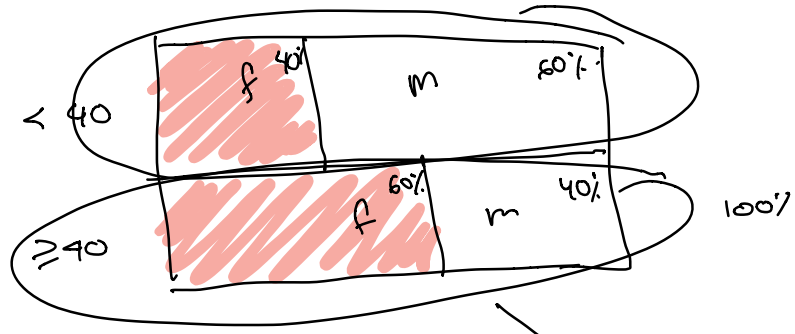
$$P(A \text{ and } B) = P(B)P(A \mid B) = P(A)P(B \mid A)$$

The *law of total probability* Let $B_1, B_2, \ldots B_n$ n sets that partition (i.e $1 = \sum_{i=1}^{n} P(B_i)$) the sample space

$$P(A) = \sum_{i=1}^{n} P(A \text{ and } B_i) = \sum_{i=1}^{n} P(B_i)P(A \mid B_i)$$

Conditional Probability

$P(f \text{ and } \geq 40) =$



"marginal"

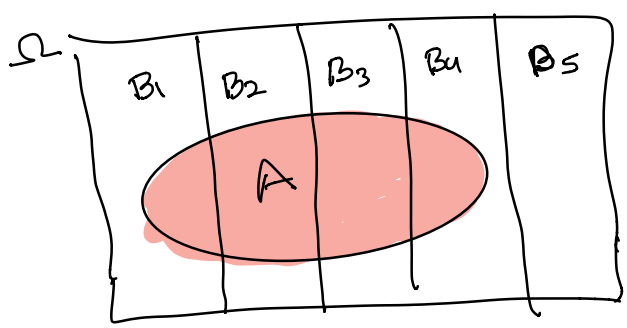
$P(f) =$

$P(f \mid \geq 40)$ ← my new population

↑ this is now my reference population

$P(f \mid \geq 40) = 0.6$

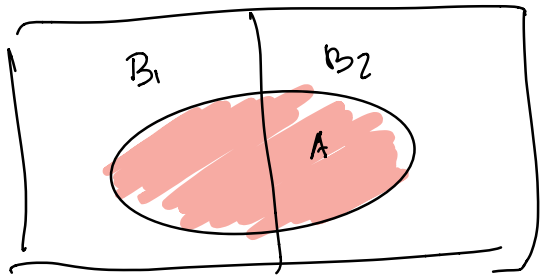
The law of total probability



$B_1, B_2 \ldots, B_5$ partition Ω

$$P(A) = P(A \text{ and } B_1) + P(A \text{ and } B_2) + P(A \text{ and } B_3)$$

$+ P(A \text{ and } B_4) + P(A \text{ and } B_5)$

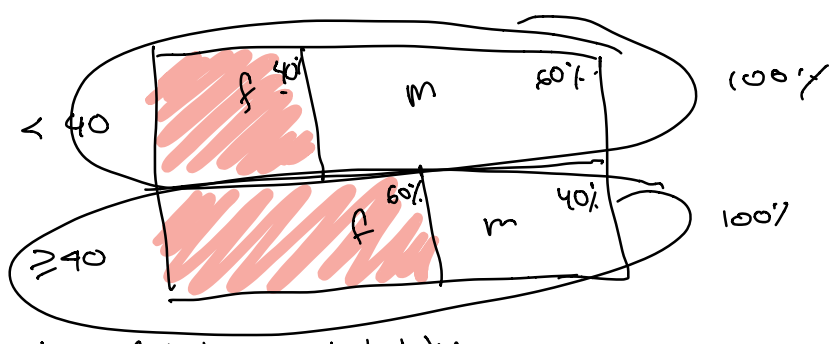


$$P(A) = P(A \text{ and } B_1) + P(A \text{ and } B_2)$$

$P(A \text{ and } B) = P(A)P(B)$ if independent.

In the general case

$$P(A \text{ and } B) = P(A)P(B|A)$$
$$= P(B)P(A|B)$$



law of total probability

$$P(f) = P(f \text{ and } < 40) + P(f \text{ and } \geq 40)$$

Bayes' rule

$$= P(f \mid < 40)P(< 40) + P(f \mid \geq 40)P(\geq 40)$$
$$= (0.4)(0.5) + (0.6)(0.5)$$

5.2 Sampling without replacement

One common use of conditional probability is sampling without replacement. That is when the occurrence of a specific outcome eliminates that outcome from the set of possibilities in subsequent random trials.

For example consider drawing cards randomly from a card deck (52 cards) shuffled. What is the probability of drawing three cards in the precise sequence "A-A-2"? $P(\text{first is A}) = \frac{4}{52}$ $P(\text{2nd is A}) = \frac{3}{51}$ $P(\text{3rd is a 2}) = \frac{4}{50}$

In contrast, when sampling with replacement, the sampled individual is not removed from the population.

$P(\text{first A}) = 4/52$ $P(\text{2nd is A}) = 4/52$ $P(\text{3rd is 2}) = 4/52$

When sampling populations for biological study, we usually choose populations that are large enough that the sampling of each individual does not change the probability distribution of possible values in the individuals that remain.

5.3 Random variable

A **random variable** is simply a variable that takes on *numerical values* that depend on the outcome of a chance operation (random experiment).

Example. Consider the experiment to tossing a die. Let the random variable Y represent the number of spots showing. The possible values of Y are 1,..,6. We do not know the value of Y until we have tossed the die. We can then compute

fixed value $\rightarrow y$

$$P(Y = 4) = 1/6$$

and

$$P(2 \leq Y \leq 4) = 1/2$$

Random variables come in two flavors: Discrete and Continuous.

5.4 Expected value (Mean)

The **mean** of a discrete random variable is:

$$\bar{Y} = E(Y) = \sum_{i=1}^{n} y_i P(Y = y_i)$$

where y_i's are the values that the variable takes on and the sum is taken over all n possible values that Y can take.

Example: Calculate the expected value of rolling a fair die once.

1 1 2 2 3 3 3 3

$$\frac{\overbrace{1+1}^{2}+\overbrace{2+2}^{2}+\overbrace{3+3+3+3}^{4}}{8}$$

$$1\left(\frac{2}{8}\right)+2\left(\frac{2}{8}\right)+3\left(\frac{4}{8}\right)$$

all possible diff. values

$$\sum_{i=1}^{n} y_i \, P(Y=y_i)$$

Example: Calculate the expected value of rolling a fair die once.

Y = value of die face

$$E[Y] = \sum_{i=1}^{6} i \, P(Y=i)$$

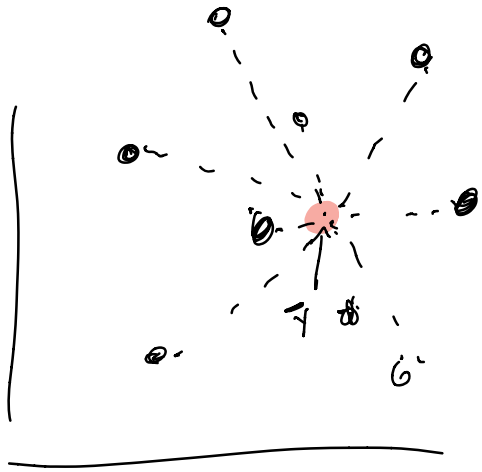
$$= \sum_{y \in \{1,\dots,6\}} y \, P(Y=y)$$

$$= (1)\left(\frac{1}{6}\right)+2\left(\frac{1}{6}\right)+3\left(\frac{1}{6}\right)+4\left(\frac{1}{6}\right)+5\left(\frac{1}{6}\right)+6\left(\frac{1}{6}\right)$$

$$= \frac{1}{6}\left[\underbrace{1+2+3+4+5+6}\right]$$

$$\sum_{i=1}^{n} i = \frac{n(n+1)}{2} \qquad = \frac{1}{6}\left[\frac{6\cdot 7}{2}\right] = 3.5$$

$$Var(Y) = E[(Y-\bar{Y})^2] = E[(Y-E[Y])^2]$$



minimizes the sum of distance to all points.

Rules: If X and Y are two random variables and Z=X+Y

$$E(Z) = E(X+Y) = E(X) + E(Y)$$

$$E(aX+b) = aE(X) + b$$

5.5 Variance

The **variance** of a discrete random variable is

$$\cancel{Var(X)} = \sum_{i=1}^{n} (y_i - E(Y))^2 P(Y = y_i)$$

$$Var(Y) = E((Y - E(Y))^2) = E[Y^2] - E[Y]^2$$

5.6 Discrete probability models

In many contexts-such as molecular biology-we count events (e.g., codons, DNA reads, CG diagrams), producing discrete variables, unlike continuous measures such as mass or intensity. Some variables are binary (e.g., mutation: yes/no), while others have multiple levels (e.g., genotypes AA, Aa, aa) or even many categories (e.g., bacterial types or 64 codons).

To summarize categorical data, we tally the frequency of each level. In R, these are stored as factors, which make tabulation straightforward. For example, here we capture the different blood genotypes for 19 subjects in a vector which we tabulate.

```
genotype = c("AA","A0","BB","A0","00","A0","AA","B0","B0",
             "A0","BB","A0","B0","AB","00","AB","BB","A0","A0")
table(genotype)
```

```
genotype
AA  AB  A0  BB  B0  00
 2   2   7   3   3   2
```

If the order in which the data are observed does not matter, we call the random variable **exchangeable**. In this case, the vector of frequencies is sufficient to capture all the relevant information in the data, providing an efficient way to summarize it.

5.6.1 Bernoulli trials

The **Bernoulli distribution** is the simplest probability model, based on a so-called Bernoulli random variable, which can take only two outcomes: failure or success. The key idea is that you have a parameter p, which represents the probability of success, and the complementary event is called probability of failure $1 - p$:

Probability distribution:

Extra activity +5 pts on Quiz 2

```
x <- rep(0, 1000)
y <- c(1, 2, 3, 4, 5, 6)
for (i in 1:1000){
  x[i] <- sample(y, 1)
}
```

x <- matrix (0, ncol=1
nrow=1000)

mean(x)

variance(x)

- $E[x^2] - E[x]^2$
- mean $((x - E(x))^2)$